Speech Communication 176 (2026) 103345

Contents lists available at ScienceDirect

Speech Communication

journal homepage: www.elsevier.com/locate/specom

Lateral channel dynamics and F3 modulation: Quantifying para-sagittal articulation in Australian English /l/

Jia Ying

School of Humanities and Social Science, The Chinese University of Hong Kong, Shenzhen, Guangdong 518172, China

ARTICLE INFO

Keywords:
/l/ production
Articulation and acoustics
Tongue lateralization
Electromagnetic articulography

ABSTRACT

This study investigates articulatory-acoustic relationships in Australian English /l/ using simultaneous 3D electromagnetic articulography (EMA) and acoustic recordings from six speakers producing /l/ in onset and coda positions with /æ/ and /ɪ/ vowels. Linear mixed-effects models revealed significant relationships between tongue lateralization and all three formants, with F3 emerging as the primary acoustic correlate of lateralization ($\beta = 0.081$, $p < 0.001$). Acoustic properties of /l/ were strongly influenced by vowel context, with significant vowel-lateralization interactions for F1 and F2, indicating that the acoustic consequences of lateralization vary by vowel environment. Temporal analysis revealed position-dependent timing relationships: F3 preceded articulatory peaks in coda position but showed near-synchronous timing in onset position, while F1 and F2 consistently lagged behind articulatory peaks across all conditions. These findings suggest distinct articulatory-acoustic coupling mechanisms for onset versus coda /l/, with F3 serving as an anticipatory cue in coda position. The results highlight the complex, context-dependent nature of /l/'s articulatory-acoustic relationships and underscore the importance of considering both spectral and temporal dimensions in understanding liquid consonant production.

1. Introduction

The lateral approximant /l/ is a segment characterized by considerable articulatory and acoustic variability across English varieties, influenced by factors such as syllable position, vowel context, and dialectal differences (Huffman, 1990; Cruttenden, 2008; Turton, 2017). This variability is particularly evident in certain varieties of English, where syllable-onset positions, commonly referred to as light [l], exhibit distinct articulatory configurations such as tongue tip raising, while syllable-coda positions, often termed dark [ɫ], are characterized by dorsum retraction and pharyngealization (Giles and Moll, 1975; Sproat and Fujimura, 1993; Browman and Goldstein, 1995; West, 1999; Wrench and Scobbie, 2003; Ladefoged, 2001; Lin, 2011; Proctor, 2011). These positional differences in /l/ production highlight the dynamic interplay of articulatory kinematics that underlie its acoustic variability.

While the mid-sagittal articulatory kinematics of /l/ have been extensively documented, the para-sagittal kinematics associated with lateral channel formation are still under-characterized. Para-sagittal movements refer to the symmetrical or asymmetrical lowering of the tongue blade to facilitate airflow along the sides of the oral cavity, creating lateral channels that are hypothesized to shape the acoustic properties of /l/ (Narayanan et al., 1999; Proctor, 2011). However, much of the existing literature emphasizes static mid-sagittal configurations or isolated coronal snapshots, yielding limited insight into the spatiotemporal formation and variability of these lateral channels (Wrench and Scobbie, 2003; Lin, 2011; Turton, 2017).

Recent advances in imaging technologies, such as magnetic resonance imaging (MRI), three-dimensional/four-dimensional (3D/4D) ultrasound imaging, and three-dimensional electromagnetic articulography (3D EMA), have enabled more detailed investigations of tongue movements in both sagittal and coronal planes (Charles and Lulich, 2018, 2019; Hwang et al., 2019; Hwang et al., 2019; Ying et al., 2021; Szalay et al., 2024). For example, studies of lateral /l/ using 3D/4D ultrasound have demonstrated asymmetrical tongue configuration in Brazilian Portuguese (Charles and Lulich, 2018, 2019), while volumetric MRI data from Szalay et al. (2024) revealed both asymmetrical tongue configurations and active lateral channel formation in Standard Southern British English (SSBE).

Despite these developments, the articulatory and acoustic consequences of lateralization remain poorly understood, particularly in dynamic contexts such as connected speech. The present study investigates lateral-channel formation and its acoustic consequences for /l/ in

E-mail address: yingjia@cuhk.edu.cn.

https://doi.org/10.1016/j.specom.2025.103345
Received 2 March 2025; Received in revised form 2 December 2025; Accepted 12 December 2025
Available online 13 December 2025

Australian English using 3D EMA, providing new empirical evidence on lateralization dynamics in this variety.

To avoid ambiguity in what follows, "gesture" is used in the Articulatory Phonology (AP) sense to denote abstract constriction-goal units, whereas "movement" refers to the observed kinematic trajectories of articulators.

1.1. Articulatory characteristics of /l/

The articulation of /l/ involves a complex interplay of mid-sagittal and para-sagittal tongue movements, which together shape its production and acoustic properties. These articulatory characteristics can be analyzed along two dimensions: (a) mid-sagittal dynamics, including the interplay of tip and dorsum movements as well as the distinction between clear and dark variants, and (b) para-sagittal dynamics, involving the formation of airflow channels along the sides of the tongue. Considering both dimensions highlights the limitations of exclusively mid-sagittal analyses and underscore the need for a more comprehensive investigation of lateralization.

The mid-sagittal profile of /l/ traditionally shows alveolo-palatal contact, while the sound's characteristic lateral airflow occurs along one or both sides of the tongue (Ladefoged and Maddieson, 1996). The physical movements of tongue-tip raising/fronting and tongue-dorsum lowering/retraction vary in prominence by syllable position, realizing the abstract AP gestures posited in phonological accounts. Onset /l/ typically emphasizes the tip gesture, while coda /l/ favors the dorsum gesture (Sproat and Fujimura, 1993; Browman and Goldstein, 1995). Imaging studies, such as ultrasound and EMA, confirm these patterns across dialects. For example, Lin (2011), Turton (2017), and Lim et al. (2019) demonstrate that syllable position affects both the timing and magnitude of tongue gestures. Dark [ɫ] variants, common in American English (AmE) and British English (BrE) coda positions, are characterized by a retracted dorsum and narrowed upper pharynx, with variable anterior tongue contact. Cross-linguistic and L2 research also shows that advanced Korean learners of English produce word-final dark [ɫ] with a retracted dorsum and lowered body, distinct from their Korean clear [l], approximating native AmE strategies (Hwang et al., 2019). These articulatory features have been widely studied, providing a foundation for understanding positional variability in /l/ production.

However, while mid-sagittal views capture critical aspects of /l/ articulation, they are insufficient to explain the airflow dynamics and tongue flexibility involved in lateralization (Alwan et al., 1997; Narayanan et al., 1997; Stone et al., 1991; Stone et al., 1992; Stone and Lundberg, 1996; Campbell and Gick, 2003; Proctor, 2011). Lateralization refers to the formation of airflow channels along the sides of the tongue, which are actively shaped by para-sagittal tongue blade gestures (Proctor, 2011; Ying et al., 2021; Szalay et al., 2024). These channels vary in symmetry and geometry depending on speaker-specific and contextual factors (Narayanan et al., 1999; Gick et al., 2017; Charles and Lulich, 2019). Unlike mid-sagittal dynamics, para-sagittal kinematics provide unique insights into the three-dimensional nature of /l/ articulation.

Lateral channel formation has been inferred from imaging techniques such as MRI, volumetric MRI, 3D EMA, and ultrasound, which highlight the independent movements of the tongue blade, tip, and dorsum. For example, Ying et al. (2021) used 3D EMA to show asymmetrical tongue shapes in AusE /l/, where one side lowers to dominate airflow while the other rises, potentially forming a side branch cavity. Similarly, Szalay et al. (2024), using volumetric MRI, demonstrate asymmetrical tongue configurations and active lateral channel formation in SSBE /l/, particularly in front-vowel contexts where coarticulation disrupts consistent tongue configurations.

Dynamic multi-modal studies reveal how lateral channel formation evolves during connected speech. Co-registered 3D EMA and ultrasound investigations by Strycharczuk et al. (2020) tracked lateralization dynamics in New Zealand English /l/, showing that lateralization peaks midway through vowel-lateral sequences. However, the degree of lateral channel formation varies significantly across /l/ variants: dark /l/ shows reduced lateral channel formation due to increased dorsal retraction, while vocalized variants exhibit minimal lateralization with limited asymmetrical tongue lowering. Crucially, their reveals that de-lateralization occurs gradually, with the lateral channel progressively narrowing before apical contact is lost entirely in vocalized forms. This suggests that vocalization emerges through progressive reduction of lateralization rather than sudden articulatory reorganization. Vowel context further modulates lateral channel dynamics, with following vowels /ɪ, ɔː/ inducing greater reduction in lateralization compared to /iː/, as demonstrated by Strycharczuk et al. (2020). These lateralization patterns fundamentally shape both the aerodynamic properties of /l/ production and its acoustic realization. To advance understanding of this relationship, the following section considers both articulatory and acoustic dynamics.

1.2. Articulatory and acoustic dynamics of /l/

The acoustics of /l/ are shaped by two key articulatory mechanisms: mid-sagittal articulation and para-sagittal lateral channels. The mid-sagittal articulation influences the back cavity and produces a low-frequency, Helmholtz-like resonance effect, where the narrow apical constriction acts like a bottle neck. Conversely, the para-sagittal lateral channels contribute to spectral zeros, also known as anti-resonances. Zeros are frequencies at which sound energy is minimized or canceled due to destructive interference between acoustic pathway. These zeros create spectral discontinuities, which are abrupt drops in energy at specific frequencies, that can severely attenuate expected formants and create gaps in the spectrum. For /l/, this interference occurs between the lateral channels and central oral cavity. This section explores (1) the core formant patterns of /l/, and (2) how lateral channels and zeros shape its spectral properties. Together, these articulatory mechanisms account for the distinctive spectral profile of /l/ and its variability across vowel context, syllable position, speaker, and dialect. These acoustic phenomena and their variation in different /l/ allophones are visually demonstrated in Fig. 1, which illustrate the distinct characteristics of the approximant in *led* and *yell* (adapted from Ladefoged and Johnson, 2014, p. 213).

The formant structure of /l/, particularly F1 and F2, reflects key articulatory configurations and provides important acoustic markers of its production. F1 is typically low (~250–500 Hz), reflecting the

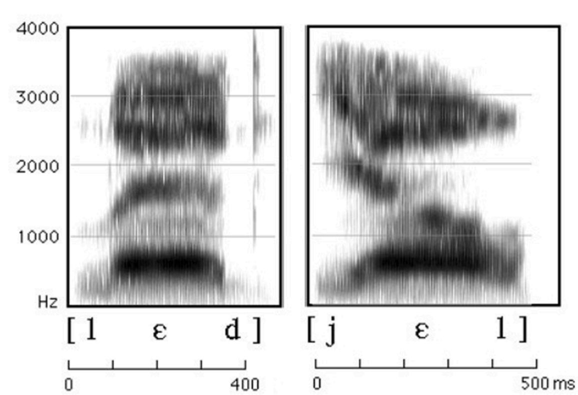

Fig. 1. Cropped spectrogram of the approximants in *led* and *yell*, showing the acoustic differences between light [l] and dark [ɫ]. The panels illustrate strong low-frequency formant energy and the attenuation of higher frequencies due to lateral anti-resonances. A higher F2 is visible in the panel for light [l], while the panel for dark [ɫ] shows a lower F2 and more energy concentrated in the low frequencies. Adapted from A Course in Phonetics (7th ed., p. 213), by P. Ladefoged & K. Johnson, 2014, Cengage Learning. Copyright 2014 by Cengage Learning.

Helmholtz-like resonance of the back cavity formed by lateral constriction (Fant, 1960; Stevens, 1998; Espy-Wilson, 1992). This low frequency is consistent across dialects and speaker groups (Lehiste, 1964; Bladon and Al-Bamerni, 1976; Turton, 2017). Theoretical models suggest that lateralization reduces the vocal tract's cross-sectional area, lowering F1, while reduced lateralization expands the area, raising F1 (Stevens, 1998). F2 is primarily influenced by tongue-body fronting and retraction. It is typically higher in onset /l/ due to a fronted dorsum and lower in coda /l/ due to dorsum retraction (Lehiste, 1964; Bladon and Al-Bamerni, 1976; Turton, 2017). Browman and Goldstein (1995) emphasize the role of dorsum retraction in altering the resonance properties of the back cavity, making F2 a key marker of darkness.

The acoustic properties of F3 in /l/ are shaped by the interaction between anterior cavity morphology and the side-branch zeros that arise from lateral channels. While F3 is often elevated in /l/ due to anterior cavity, side-branch zeros resulting from lateral channel asymmetry and supralingual cavity configurations can attenuate or mask energy in this region (Zhang and Espy-Wilson, 2004). For instance, asymmetric lateral openings can introduce a zero that attenuates energy around 2000–5000 Hz, depending on channel geometry. Stevens (1998) further linked F3 (∼2800 Hz) to anterior cavity shaping and side-branch geometry in refined models, estimating shorter side branches (2–4 cm) and smaller volumes (2–3 cm^3). However, the dynamic and spatiotemporal relationship between lateralization and F3 has not been extensively documented, meaning the full influence of side-branch geometry on the resulting spectral profile, including the effects of higher anti-resonances and zeros, remains to be fully explored.

The spectral shape of /l/ is characterized by prominent low formants and attenuation of higher formants due to anti-resonances and zeros (Fant, 1960; Stevens, 1998; Espy-Wilson, 1992; Narayanan et al., 1997). As established earlier, these zeros result from destructive interference between sound radiating through the lateral channels and the central oral cavity, creating frequency-specific energy nulls. The exact frequencies and attenuation magnitudes of these zeros are influenced by lateral channel geometry, asymmetry, and the interaction between mid-sagittal and para-sagittal articulatory movements (Narayanan et al., 1999). Understanding these spectral features requires consideration of the articulatory mechanisms that give rise to them.

Recent studies reveal the importance of lateral asymmetry and para-sagittal tongue blade movements in shaping /l/ acoustics. Narayanan et al. (1997) confirmed asymmetrical channels using MRI and EPG, linking low F1 (250–500 Hz) to back-cavity resonance and F2 (1250–1450 Hz) to cavity length. Zhang and Espy-Wilson (2004) demonstrated that asymmetrical lateral openings introduce zeros sensitive to branch length and area variations, affecting F2 and F3 frequencies. Charles and Lulich (2018, 2019) found that Brazilian Portuguese /l/ exhibits prominent anti-formants (∼2000–5000 Hz) arising from lateral channels and supralingual cavities, with additional variability tied to symmetrical vs. asymmetrical articulations. Building on these findings, Szalay et al. (2024) used rtMRI and volumetric MRI to show that lateral channel formation and supralingual cavity configuration produce anti-formants and reduced intensity in /l/ across vowel context. These findings establish lateral channel geometry and supralingual cavity configuration as key determinants of /l/ acoustic variability. Advanced MRI techniques now offer direct evidence linking these articulatory features to anti-formant production and reduced intensity.

Computational acoustic modeling further clarifies how the physical structure of lateral channels and supralingual cavities shapes the spectral properties of /l/, complementing imaging-based findings. Acoustic modeling of /l/ using VTAR (Vocal Tract Acoustic Response; Zhang and Espy-Wilson, 2004), a MATLAB-based tool that computes the vocal tract's frequency-domain acoustic response, reveals pole–zero clusters between 2–5 kHz shaped by lateral channels and supralingual cavities. Simulations demonstrate how varying channel lengths and areas impact spectral shape, validating the role of airflow patterns in shaping /l/ acoustics. MRI-derived models similarly highlight zeros arising from lateral channels and supralingual branches, emphasizing the importance of para-sagittal airflow dynamics. However, methodological differences such as reliance on sustained vowel productions rather than dynamic vowel-lateral transitions, along with limitations like single-speaker designs and technical constraints of rtMRI and in-scanner audio, may limit the generalizability and precision of previous findings.

1.3. *The present study*

The present study investigates the dynamic relationships between the articulatory and acoustic properties of lateral channel formation. Using 3D EMA synchronized with acoustic recordings, this study tracks mid-sagittal and para-sagittal tongue movements in six speakers to examine how lateralization evolves over time and influences formant trajectories. 3D EMA is particularly suited for this purpose due to its ability to capture real-time kinematic data with high spatial and temporal precision (Byrd, 2000). By tracking the movements of multiple articulators, EMA enables a detailed analysis of lateral channel formation and para-sagittal contributions, providing the necessary insights into the complex articulatory kinematics essential for understanding lateralization in speech production. To guide the investigation, four hypotheses are presented based on prior articulatory and acoustic studies of /l/:

Hypothesis 1: Tongue lateralization's effect on F1 frequency during the /l/ transition varies with the preceding vowel's articulatory configuration. a) For /æ/: the large articulatory distance between the low vowel and high /l/ target triggers a compensatory strategy during lateralization where the tongue body raises to facilitate the transition, resulting in F1 decrease. b) For /ɪ/: lateralization involves the expected blade lowering, resulting in F1 increase. These context-dependent effects reflect the interaction between lateralization and the distinct articulatory configurations of the preceding vowels (Fant, 1960; Stevens, 1998).

Hypothesis 2: Unlike F1, which directly reflects changes in vocal tract constriction degree, F2 frequency is expected to show weaker correlations with tongue lateralization. F2 is primarily determined by tongue dorsum advancement/retraction and oral cavity length (Fant, 1960), while lateralization specifically measures asymmetric lowering of the tongue blade sides. Since tongue dorsum positioning can vary independently of blade lateralization during /l/ production (Browman and Goldstein, 1995), the relationship between lateralization and F2 is predicted to be minimal. This articulatory independence between different tongue regions during lateral production has been documented in previous studies (Lin and Demuth, 2015).

Hypothesis 3: Increased tongue lateralization will correlate with systematic changes in F3 frequency trajectories. While the literature establishes that lateral channels create zeros and modify cavity resonances, the direction of F3 frequency shift remains an empirical question. Two competing mechanisms could apply: (a) the rising of the non-dominant tongue side could raise F3 by reducing the supralingual cavity size, or (b) expanded lateral openings could lower F3 through altered acoustic coupling. Studies have linked lateral tongue configurations to F3-related acoustic phenomena (Zhang and Espy-Wilson, 2004; Charles and Lulich, 2019; Szalay et al., 2024).

Hypothesis 4: While syllable position strongly affects the relative timing of tongue tip and tongue body gestures in /l/ production (Sproat and Fujimura, 1993), the degree of tongue lateralization itself remains relatively stable across onset and coda positions (Ying et al., 2021). Since F1 and F3 are predicted to correlate specifically with lateralization degree (as opposed to other aspects of /l/ articulation), these formants should show minimal variation between syllable positions. This prediction distinguishes lateralization-related acoustic effects from the well-documented positional variations in tongue body retraction and timing, which primarily affect F2.

2. Methods

This study examined articulatory-acoustic relationships in lateral channel formation using the Australian English /l/ production dataset from Ying et al. (2021). The articulatory measurements and lateralization indices from that study were paired with time-aligned acoustic recordings to investigate how lateral channel formation affects formant structure.

2.1. Participants

Six monolingual AusE speakers (three females and three males; mean age 22.2 years, range = 19-35 years) participated in the study. None of the participants had atypical speech, and none demonstrated pervasive syllable-final lateral vocalization; that is, they articulated their final /l/s as lateral approximants rather than as high back vowels. All participants were living in Sydney at the time of data collection. They were paid for their participation and were naïve to the purpose of the experiment.

2.2. Experimental material

Laterals were elicited word-medially in disyllabic words of the form /ˈCVb.lət/ or /ˈ(C)Vl.bət/, allowing comparison of both syllable-onset and syllable-coda /l/s. In both syllable-onset and syllable-coda positions, laterals were preceded by a stressed front vowel, either /æ/ or /ɪ/. The vowels were chosen because of the different constraints that they place on the shape of the tongue preceding /l/. Prior analysis of these articulatory patterns demonstrated differential vowel effects on /l/ articulation, with significant mid-sagittal modifications but minimal para-sagittal adjustments across phonetic contexts (Ying et al., 2021).

Target words were read aloud in the carrier phrase ‘keep _ here’, with adjacent /p/ and /h/ chosen to minimize lingual coarticulation effects. /b/ preceding or following the /l/ is used in both target forms for this purpose as well (see Table 1). The stimuli were presented in 10-word blocks, with ten repetitions for each target word, randomized across blocks for each participant. Each recording session took approximately 25 minutes for a participant to complete. 516 valid tokens from six participants were used for analysis in this study.

2.3. Procedure

Experiments were conducted at the MARCS Institute Speech Production Laboratory at Western Sydney University. Articulographic data were acquired at a rate of 100 Hz using an NDI Wave electromagnetic articulography (EMA) system (Northern Digital Inc., Canada). Synchronized companion speech audio was recorded at a 22,050 Hz sampling rate using a Schoeps Colette Series Supercardioid microphone and EURORACK UB802 preamplifier. Tongue, lip and jaw movements were tracked using three EMA sensors affixed mid-sagittally to the tongue tip (TT; ~5 mm behind the apex), tongue middle (TM; ~20 mm behind the TT sensor) and tongue dorsum (TD; between 20 and 35 mm behind the TM sensor, depending on the participant’s tolerance), and another two sensors affixed para-sagitally to the sides of the tongue blade (on the top surface ~5 mm from the edges of the tongue and ~15 mm from both the TT sensor and the TM sensor). The TD sensor was located 45 to 60 mm posterior to the TT sensor, depending on each speakers’ comfort level.

Table 1
Target words used to elicit Australian English (AusE) /l/s. Note that AusE is a non-rhotic variety of English, thus final ‘-ert’ is pronounced [ət].

PRECEDING VOWEL	æ		ɪ	
MID-SAGITTAL TONGUE SHAPE	concave		convex	
REPRESENTATION	ORTHOGRAPHIC	IPA	ORTHOGRAPHIC	IPA
CVC.lVC	tablet	[ˈtæb.lət]	tiblet	[ˈtɪb.lət]
(onset /l/)	cablet	[ˈkæb.lət]	kiblet	[ˈkɪb.lət]
(C)Vl.CVC	Talbot	[ˈtæl.bət]	Tilbert	[ˈtɪl.bət]
(coda /l/)	Calbert	[ˈkæl.bət]	Kilbert	[ˈkɪl.bət]
	Albert	[ˈæl.bət]	Ilbert	[ˈɪl.bət]

Partly due to previous limitations in sensing technology, and partly because of the focus on mid-sagittal articulation in the existing literature, there has been comparatively little prior research on para-sagittal dynamics of tongue movements in speech. To address this gap, the ‘Southern Cross’ sensor configuration was developed, incorporating para-sagittal sensors on each side of the tongue blade to enable measurement of para-sagittal dynamics (Ying et al., 2021). Fig. 2 provides a schematic of tongue sensor placement for this configuration.

Sensors were also attached to the lower jaw on the gum line between the two central incisors; to the upper lip and lower lip along the vermillion border in the mid-sagittal plane; to the left mastoid (LM) and right mastoid (RM); and to the nasion (NA). The LM, RM and NA sensors were used for correction of head motion for post-collection data processing. Three sensors are required to account for the translation and rotation of the head using x, y, and z coordinates. The occlusal (bite) plane was determined by having speakers clench a semi-circular protractor between their upper and lower teeth. Two sensors were attached to the corners of the protractor and the third sensor was attached to the center of the circular portion of the protractor to define a rigid occlusal plane.

Participants were familiarized with the target words before recording. Elicitation sentences were presented on a computer monitor placed approximately 120 cm in front of the participant, and participants were instructed to read the sentences at a comfortable speaking rate. Participants may be slightly irritated by having the sensors placed on the tongue. However, they usually adjusted to the situation quickly. Thus far no comparative analysis has shown that the articulation is significantly affected by the sensors (Katz et al., 2006; Gibert et al., 2015).

2.4. Data processing and measurements

Articulographic data were corrected for head movement and rotated into a common coordinate system: x = anterior-posterior; y = left-right; z = up-down. Sensor displacement was expressed with respect to an origin located on the speaker’s occlusal plane, along the midline and immediately behind the upper incisors. Kinematic data from the lingual sensors were filtered and smoothed using a robust DCT-based penalized least squares algorithm (Garcia, 2010). Smoothing splines (gss R package; Gu 2002) were applied to time-varying measurements derived from the kinematic data to observe general trends across tokens.

EMA data were first visualized using MVIEW, a MATLAB-based program developed by Mark Tiede at Haskins Laboratories (Tiede, 2005). MVIEW displays the positional signal of the sensors, time-aligned with the acoustic speech signal. Visualization of the data revealed that /l/ production primarily involved horizontal (x) motion of the TD sensor, and vertical (z) motion of the TM, TT and two para-sagittal tongue blade sensors, tongue blade left (TL) and tongue blade right (TR).

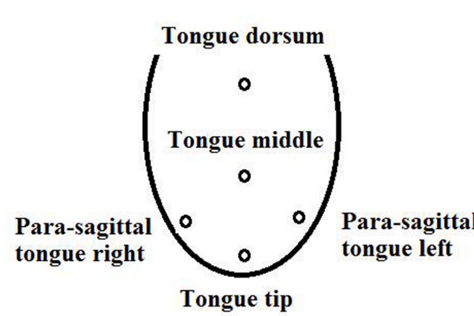

Fig. 2. Tongue sensor positions in the ‘Southern Cross’ configuration designed for articulographic investigation of mid-sagittal and para-sagittal movements in speech.

A set of temporal landmarks was identified in the acoustic signal to define the window in which the articulatory data would be measured. These landmarks were identified by visual inspection of acoustic waveforms and spectra in Praat (Boersma and Weenink, 2015), and articulatory analysis was subsequently conducted in MATLAB with custom scripts. In words with syllable onset-/l/, the TTz extremum (maximum of vertical displacement, i.e., highest position, or peak) occurred at the acoustic onset of the /ə/, as in /ˈ(C)Vb.lət/. In words with syllable coda-/l/, the extremum of TTz typically occurred at the acoustic offset of the /b/, as in /ˈ(C)Vl.bət/. Based on these observations, Vb.l segment sequences for onset-/l/ words and Vl.b segment sequences for coda-/l/ words were both demarcated in Praat using the acoustic onset of stressed pre-lateral vowel and the acoustic onset of the unstressed post-lateral vowel as the boundary endpoints of the analysis window. This segmentation protocol ensured that the TTz gesture extremum associated with /l/ production would occur within the segmentation boundaries for both coda and onset /l/ tokens. Fig. 3 shows two examples of these acoustic landmarks in vowel-/l/ sequences produced by female speaker F03.

2.4.1. Acoustic measurements

Noise reduction was applied to all files using Audacity (Version 2.0.6). The parameters were set at 30 dB for noise reduction (dB), 1 dB for sensitivity (dB), 200 Hz for frequency smoothing (Hz), and 0.15 seconds for decay time (seconds). After reducing the noise, a three-formant estimation (F1, F2, and F3 frequency) was conducted in Praat using the Burg linear predictive coding (LPC) method. The F3 ceiling, which varied across speakers (3600, 3400 and 3900 Hz for the females, but 3200 Hz for all three males), was used to optimize formant estimation independently for each speaker, similar to Escudero et al. (2009). These F3 ceiling heights were automatically determined for each speaker in the following way. First, F1-F3 measurements were taken with the F3 ceiling increasing iteratively in 50 Hz steps, from 2000 to 4000 Hz. The speaker-dependent measurements were imported to R, where the variance for each formant was calculated at each step, and the three formant variances were summed together. The summed variances were plotted, and the F3 ceiling that resulted in the lowest variance was logged (i.e., the F3 frequency ceiling that resulted in the most consistent measurements of F1, F2, and F3 frequency for the given speaker's data). The speaker-optimized ceilings were verified manually in Praat by comparing the resulting formant tracks against a broadband spectrogram. The formant tracks based on the speaker-optimized F3 ceiling frequencies were generated throughout the same analysis window shown in Fig. 3.

2.4.2. Articulatory measurements

Temporal alignment of TD trajectories (horizontal dimension), and TM, TT, TL and TR trajectories (vertical dimension) is illustrated in Fig. 4a and 4b, for the words tablet and talbot; differences in TT height and TD retraction are compared across syllable positions. In both syllable positions, the para-sagittal left and right sides of the tongue blade (TL and TR) are raised in concert with TT during /l/ production, with at least one side of the tongue blade (either TL or TR) slightly higher or lower than TT. For speaker F03, TL is about 1.35mm lower than TT at about 360ms in *tablet*, and about 1.5mm higher at about 270ms in *talbot*. TDx retraction and TMz lowering can be observed in both tokens. The maximum TD retraction is about -42.6mm (relative to the occlusal plane origin behind the upper front teeth) at 130ms and the maximum TM lowering is about 0.7mm at 300ms in *tablet*. In *talbot*, the maximum TD retraction is about -43.8mm at 180ms and the maximum TM lowering is about -3.3mm at 200ms. Then TT, TL and TR rise gradually for the /l/ and lower slightly for the following /ə/. Thus, in syllable-coda position (tal.bot), TT is lower, TM is lower and TD is slightly more retracted than in syllable-onset position (tab.let), in which TT is raised, TM is lowered and TD is slightly less retracted.

The present study developed a protocol for addressing para-sagittal dynamics in /l/ production and contributing to the understanding of lateral channel formation (lateralization). In order to estimate lateralization of the tongue blade in the coronal plane, a virtual mid-sagittal tongue blade sensor was required. In this study, a tongue blade sensor in the mid-sagittal plane was mathematically estimated from relationships among the para-sagittal TR and TL sensors and the mid-sagittal TT, TM and TD sensors (Fig. 5). The steps used to estimate the mid-sagittal tongue blade sensor were as follows.

Firstly, to approximate the local midsagittal tongue surface spanning the blade–dorsum region relevant for /l/, a second-order polynomial $z = ax^2 + bx + c$ was fit at each frame to the (x, z) coordinates of the three midsagittal sensors (TT, TM, TD) following established methodologies for EMA data analysis (Kalashnikova et al., 2017; Blackwood Ximenes et al., 2017). This approach was chosen to balance anatomical accuracy and computational simplicity: with three non-collinear points, a quadratic is the highest-order polynomial that can be fit by the data without arbitrary constraints. A first-order (linear) fit enforces zero curvature and can bias the interpolated height at interior x positions when TT and TD move in opposite vertical directions, while higher-order polynomials (≥3) are susceptible to overfitting and prone to

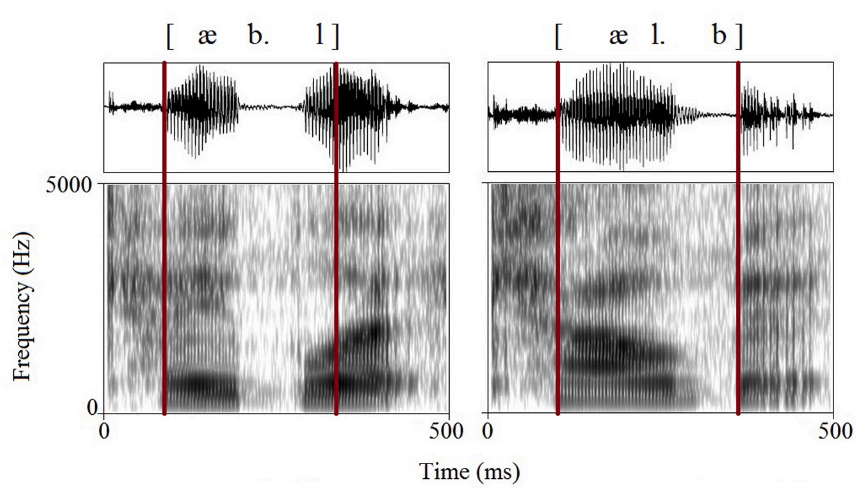

Fig. 3. (Color online) Identification of the analysis window. Acoustic waveform and spectrum of [æb.l] (left) and [æl.b] (right) produced by female speaker F03. Vertical red lines indicate the endpoint landmarks of the analysis window. Left landmark: acoustic onset of stressed pre-lateral vowel. Right landmark: acoustic onset of unstressed post-lateral vowel.

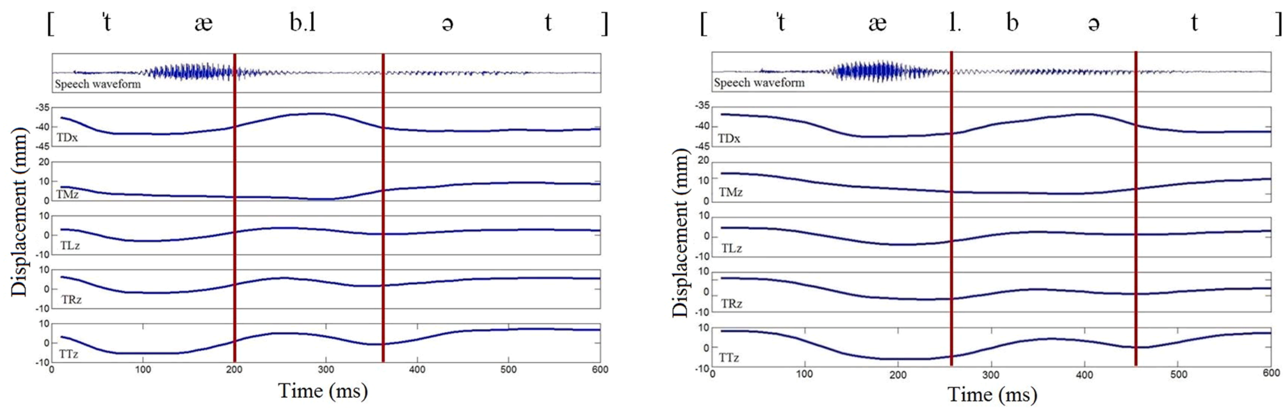

Fig. 4. (Color online) /l/ articulation in onset and coda positions. Top-to-bottom in each panel: acoustic waveform; tongue dorsum movement in the anterior-posterior (x) dimension (TDx); then tongue middle (TMz), para-sagittal tongue left (TLz), para-sagittal tongue right (TRz) and tongue tip (TTz) movements in the vertical (z) dimension. The red lines indicate the defined endpoints for /l/ articulation. Example utterances produced by female speaker F03. Panel 4a: *tablet* (onset /l/). Panel 4b: *talbot* (coda /l/). Horizontal axis: time (ms). Vertical axes: displacement (mm).

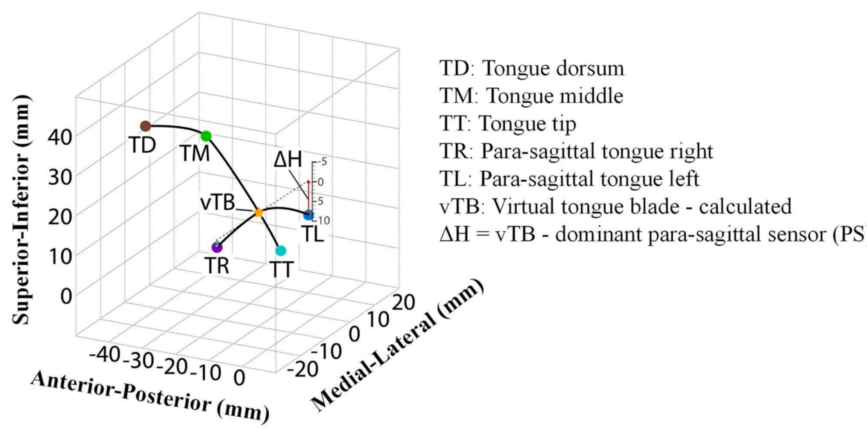

Fig. 5. Schematic illustration of a virtual tongue blade (vTB) sensor and the ΔHeight (tongue lateralization) measure projected in tongue ¾ view. The 0,0,0 point is the center of the occlusal plane (which is about 1 cm below the tongue tip). The vTB sensor (gold star) was mathematically estimated on the coronal plane perpendicular to the occlusal plane. The vTB sensor is located in the intersection of the coronal polynomial (TL – blue dot and TR – purple dot) with the sagittal polynomial (TT – turquoise dot, TM – green dot, and TD – brown dot). In order to measure tongue lateralization in the coronal plane, we created an index of the difference in height (z) between the dominant (i.e., lower) side of the tongue blade (TL in this illustration) and the mid-sagittal vTB sensor, called ΔHeight (indicated by the red arrow next to the inset scale of values). A ΔHeight value of zero indicates that the line between the dominant para-sagittal sensor and the vTB sensor is completely horizontal (i.e. flat). A positive value indicates that the line is tilted downwards (i.e. greater lateralization). A negative value indicates that the line is tilted upwards (i.e. lesser lateralization).

unstable interpolation behavior between sensor locations.

Secondly, a virtual tongue blade sensor (vTB) was defined by evaluating the fitted polynomial at a token-level time-averaged horizontal coordinate of the para-sagittal blade sensors (mean x of TR, TL). The vertical position of this virtual sensor was calculated by inputting this horizontal coordinate into the fitted second-order polynomial curve. This static (token-average) coronal plane was adopted to suppress framewise sensor jitter and lateral micro-shear (≈0.2–0.5 mm EMA noise) that a framewise plane would inject into the lateralization time series, thereby improving signal-to-noise ratio for timing estimation.

To measure tongue lateralization in the coronal plane, an index referred to as ΔHeight was created, which captures the degree to which the dominant side of the tongue blade (the lower of the two para-sagittal sensors) differed in height from the mid-sagittal vTB sensor at any given point in time. During /l/ production, the speakers in this study typically tilted one side of the tongue blade lower, either the left or right side. Formula (1) below was used to calculate the token-by-token difference between the mid-sagittal vTB sensor and the dominant para-sagittal (PS) sensor in the vertical dimension at each time sample in the analysis. Positive values reflect para-sagittal lowering relative to the interpolated midsagittal surface, indicating increased lateral channel opening.

$$\Delta Height = vTB - PS \tag{1}$$

To validate the interpolation approach, the timing measures were recomputed substituting the TT sensor directly for vTB; the inferential pattern remained unchanged, supporting that the conclusions do not depend critically on the interpolation step. Additionally, while the use of a real sensor at the tongue blade was considered, practical constraints, such as the limited space on the tongue's surface, made this option difficult to implement for participants.

2.5. Data normalization

The z-score normalization (Lobanov, 1973) method was used to standardize the articulatory and acoustic data, so that we could compare the dataset across speakers (Adank et al., 2004; Wieling et al., 2016). The data were reduced to contain only data points that were equal to or less than three standard deviations from model predictions. The y-axis in all figures of the Results section show the normalized values for ΔHeight (tongue lateralization), F1, F2 and F3 frequencies.

2.6. Statistical analysis

To evaluate the reliability of the descriptive acoustic-articulatory relationships above, three linear mixed effect (LME) models at the 300-ms interval (one each for F1, F2 and F3 relationships to ΔHeight) were fit to the normalized ΔHeight values using the lme4 package (Bates, Maechler, Bolker, Walker, 2014) in Rstudio Version 3.6.1. The results in the model summary are with treatment contrasts. The dependent variable is the formant values and the independent variable is the ΔHeight values. The vowel context (/ɪ/ and /æ) and syllable position (onset and coda) were included as fixed effects. Random intercepts were included for speaker, item (i.e., token order), and repetition of the same token in each model.

The Bonferroni-adjusted significance level was set to 0.017, for comparison across the three models. Estimates for t-statistics and p-values were generated using Satterthwaite approximation by fitting linear mixed-effects models using the lmerTest package (Kuznetsova, Brockhoff, and Christensen, 2016). Model comparison was conducted by back-fitting along the Akaike Information Criterion (AIC) to measure quality of fit. All models were conducted with a three-way interaction term (ΔHeight*vowel*syllable position). If a three-way-interaction term and a two-way-interaction term were both significant, then the three-way-interaction term and the two-way-interaction term were compared using analysis of variance (ANOVA) models with a Chi-square test to determine whether the reductions in the residual sum of squares were significant. If the two models did not differ significantly, then the simpler model was retained.

In order to examine which formant best predicts ΔHeight, relative importance analysis (RIA) was implemented through the Grömping (2006) calc.relimp package within the R statistical environment. The function calculates several relative importance metrics for linear models using a method (lmg) developed by Lindemann, Merenda and Gold (1980). lmg calculates the relative contribution of the predictor to the R^2. According to Lindeman et al. (1980, p. 119), lmg is the R^2 contribution averaged over ordering among predictors. The proportion of the variance is represented by R^2. A larger R^2 means that the predictor contributes more to variance shared with the outcome variable.

3. Results

3.1. Temporal dynamics

To examine the temporal dynamics of tongue lateralization across different phonological contexts, Smoothing Spline Analysis of Variance (SSANOVA) was employed. This statistical technique extends traditional ANOVA to functional data by fitting smooth curves to time-series trajectories, rather than comparing single time points (Davidson, 2006; Gu, 2013). This approach captures the continuous, dynamic nature of speech movements, identifying when and how trajectories differ between conditions.

SSANOVA was used to generate descriptive visualizations (Figs. 6–9) illustrating the time-course patterns of articulatory and acoustic measures across different contexts. These trajectory plots, with their confidence bands, provide clear visual representations of how tongue lateralization (ΔHeight) and formant frequencies (F1-F3) evolve throughout the /Vl/ production. While the SSANOVA figures primarily serve as descriptive illustrations to complement the quantitative analyses, they effectively reveal the temporal dynamics underlying the peak-based measurements and acoustic-articulatory relationships examined in subsequent statistical analyses.

3.1.1. Temporal dynamics of tongue lateralization (ΔHeight)

Fig. 6 the temporal trajectories of tongue lateralization (ΔHeight) across four phonological contexts: æ.onset, æ.coda, ɪ.onset, and ɪ.coda. The y-axis shows the magnitude of tongue lateralization. In this measure, a value of zero indicates that the line between the dominant parasagittal sensor and the virtual tongue blade sensor is flat, which means the tongue blade shows no lateral tilt. A positive value indicates that the dominant tongue side is lower than the tongue midline (greater lateralization), and a negative value indicates that the tongue side is higher than the midline (lesser lateralization). The x-axis shows a time window

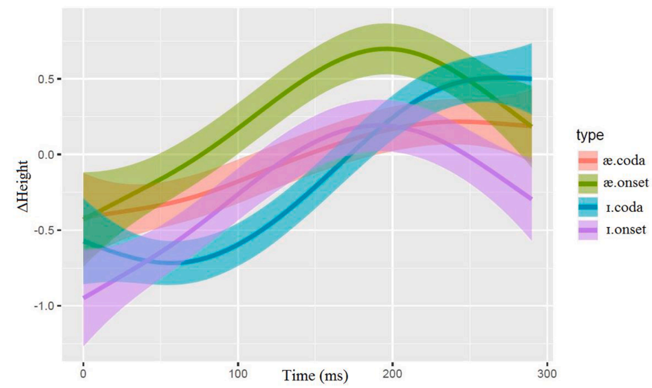

Fig. 6. (Color online) Temporal dynamics of tongue lateralization on the dominant side of the tongue blade in the coronal plane. The x-axis shows the time window of 300ms from the onset of V1. The y-axis shows normalized tongue lateralization (ΔHeight). The dominant side of the tongue goes from being higher than the virtual TB sensor (negative ΔHeight) during the preceding stressed vowel, shifts to being lower than virtual TB (positive ΔHeight: greater lateralization) by the peak of /l/, and then shifts toward flat shape going into the following unstressed vowel except for coda /l/ following both vowels.

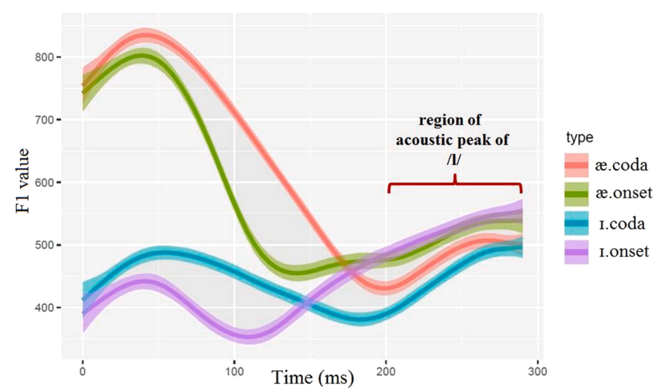

Fig. 7. (Color online) Temporal dynamics of F1 frequency for vowel x syllable position. The x-axis shows the time window of 300ms from the onset of V1. The y-axis shows F1 values. The red brace indicates the region of the acoustic peaks for /l/.

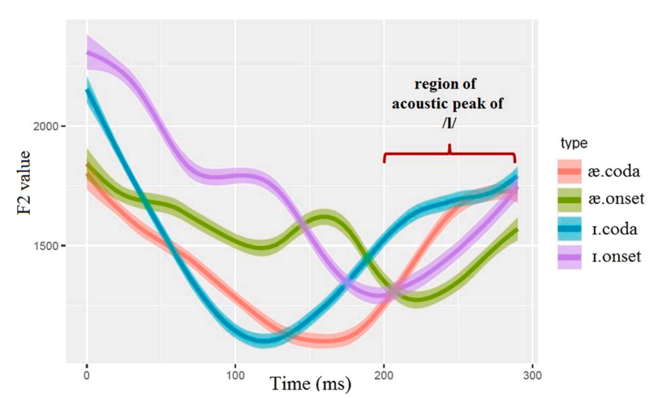

Fig. 8. (Color online) Temporal dynamics of F2 frequency for vowel x syllable position.

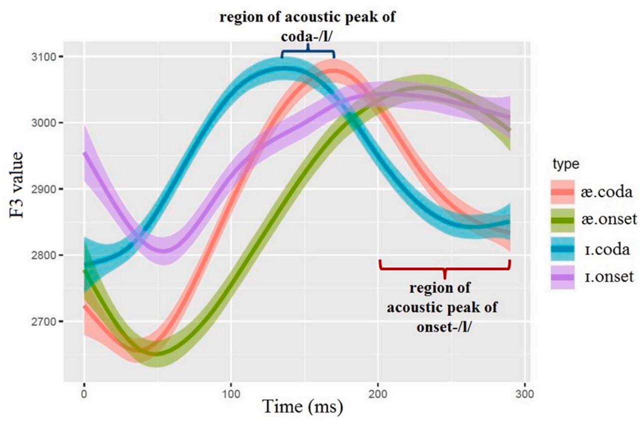

Fig. 9. (Color online) Temporal dynamics of F3 frequency for vowel x syllable position. Two regions of acoustic peaks were observed, one in red for onset /l/s, and the other in blue for coda /l/s.

of 300ms, which covers the entire /Vl/ interval. The trajectories reveal that all phonological contexts eventually reach positive lateralization peaks (dominant tongue side positioned lower than midline), but exhibit different temporal coordination patterns:

æ.onset /l/ (green) maintains predominantly positive lateralization throughout the /Vl/ production, reaching the highest peak (∼+0.6) around 200ms. This context shows sustained elevated lateralization for the majority of the production interval. **ɪ.onset** /l/ (purple) shows the most pronounced shift, beginning with strong negative lateralization (∼-1.0) where the dominant tongue side is positioned higher than midline. It then undergoes a substantial recovery to reach positive peaks (∼+0.3) around 200ms, demonstrating a complex articulatory adjustment pattern. **ɪ.coda** /l/ (cyan) demonstrates a gradual rise from initial negative values to moderate positive peaks, following a nearly linear trajectory throughout the temporal window. **æ.coda** (coral) exhibits modest fluctuations around baseline, reaching moderate positive peaks with relatively stable lateralization patterns.

3.1.2. F1 frequency

Fig. 7 shows the temporal dynamics of F1 frequency over the 300ms interval from the onset of /Vl/ production. The acoustic peak of lateralization was defined as the maximum F1 value identified within a 200-300ms window following the onset of the pre-lateral vowel. This window was chosen to align with the period of maximal tongue tip elevation as confirmed by our simultaneous articulatory data. This method allowed for the identification of the F1 maximum regardless of whether it was a sharp, discrete peak or a more sustained high value.

The trajectories reveal distinct patterns based on vowel context. The /æ/ contexts (coral and green) are characterized by pronounced F1 transitions: both æ.coda and æ.onset show the high F1 frequency of the /æ/ vowel being rapidly attenuated as articulation shifts to the low F1 of /l/, creating steep falling transitions. In contrast, the /ɪ/ contexts (purple and cyan) present markedly different patterns: both ɪ.onset and ɪ.coda maintain relatively invariant and low F1 plateaus throughout the interval, reflecting the comparable F1 frequencies between the /ɪ/ vowel and the adjacent /l/ consonant. This acoustic contrast between high-to-low transitions in /æ/ contexts versus stable low plateaus in /ɪ/ contexts represents the primary pattern in the data.

3.1.3. F2 frequency

Fig. 8 illustrates the temporal dynamics of F2 frequency over the 300ms interval. The acoustic peak for /l/ occurs after 200ms, which is later than the peak of articulatory lateralization (ΔHeight). The trajectories show distinct patterns based on both vowel context and syllable position. In coda position, ɪ.coda /l/ (cyan) begins at a high frequency and undergoes a substantial falling-rising transition, while æ.coda /l/ (coral) begins lower and shows a less pronounced initial decrease. Despite these different starting points, both coda contours converge on a nearly identical F2 frequency by the end of the interval. In onset position, the acoustic distinction between vowels is preserved differently: ɪ.onset /l/ (purple) remains consistently higher than æ.onset /l/ (green) for the majority of the trajectory, with a transient reversal at 150-200 ms where the green curve is above the purple. Although both onset trajectories display falling-rising patterns, they maintain separation, indicating that the coarticulatory influence of the preceding vowel is sustained.

3.1.4. F3 frequency

Fig. 9 shows the temporal dynamics of F3 frequency over the 300ms interval. The trajectories reveal consistent vowel-based distinctions in F3 patterns. For the coda contexts, both ɪ.coda /l/ (cyan) and æ.coda /l/ (coral) reach their F3 peaks relatively early, around 150-200ms. These trajectories show similar F3 magnitudes during the acoustic peak region, both ranging approximately 3100Hz with overlapping confidence bands. The onset contexts, æ.onset /l/ (green) and ɪ.onset /l/ (purple), demonstrate later peak timing, reaching their maxima around 250ms. Like the coda contexts, the onset trajectories converge to similar F3 values at peak, approximately 3050 Hz, with substantial overlap in their confidence intervals.

3.2. Peak time window analysis

3.2.1. Acoustic-articulatory timing: F1

Fig. 7 presents F1 dynamics across different phonological contexts. Optimized windows were used for each formant due to their specific acoustic patterns. While this approach prevents direct cross-formant statistical comparison, it ensures accurate capture of lateralization-related peaks in each acoustic dimension. Within the 200-300ms analysis window, F1peaks were identified and their timing was compared to articulatory constriction peaks using paired-samples analysis. Overall, F1 peaks occurred 13.8ms after articulatory peaks (articulatory: 240.0ms, SE = 1.8ms; F1: 254.0ms, SE = 1.3ms). This positive lag was statistically significant ($t(515) = 6.24$, $p < 0.001$, 95% CI [9.4, 18.1]), representing a small but reliable effect (Cohen's $d = 0.275$).

While the overall effect is modest, it reveals substantial condition-specific differences. The timing relationship was strongly modulated by syllable position. Onset positions, where visual differences were most salient, showed that F1 peaked considerably after articulation, with F1 for ɪ.onset peaking 28.6ms after articulation (SE = 5.1ms) and for æ.onset peaking 22.2ms after articulation (SE = 5.6ms). Conversely, coda positions displayed reduced lags, with F1 for æ.coda peaking only 8.2ms after articulation (SE = 3.7ms) and for ɪ.coda peaking 4.3ms after articulation (SE = 3.7ms). Vowel quality showed minimal influence on articulatory-acoustic lag, with both /æ/ and /ɪ/ exhibiting nearly identical mean delays (13.9ms, SE = 3.2ms vs. 13.7ms, SE = 3.1ms, respectively).

3.2.2. Acoustic-articulatory timing: F2

Fig. 8 presents F2 dynamics across conditions, and a qualitative assessment of the data suggests a substantial delay of acoustic peaks relative to lateralization peaks within the 200-300ms window. While the æ.coda /l/ shows a clear F2 peak within this window, the other three conditions (æ.onset /l/, ɪ.onset /l/, and ɪ.coda /l/) display continuous upward F2 trajectories. The extension of the analysis window to 800ms confirmed these patterns, validating the 300ms window as appropriate for analysis. For the æ.coda, F2 peaked 10.1ms after the articulatory lateralization peak (SE = 3.8ms). The continuous F2 rise in three of four conditions prevents meaningful peak-timing analysis for most of the data. Unlike F1, which showed clear peaks across all conditions, F2 peak timing could only be assessed for æ.coda.

3.2.3. Acoustic-articulatory timing: F3

Fig. 9 presents F3 dynamics, which reveal a fundamentally different temporal relationship compared to F1 and F2. A qualitative assessment of the data suggests significant timing differences between F3 and articulatory peaks, particularly for coda positions where F3 peaks appear to precede articulatory peaks by 50-75ms. This visual impression is explained with statistical reality by a more nuanced detailed analysis. Within the 100-300ms analysis window, maximum F3 values (acoustic peaks) were identified and their timing was compared to articulatory lateralization peaks using paired-samples analysis. Overall, F3 peaks occurred 11.2ms before articulatory peaks (articulatory: 196.0ms, SE = 3.3ms; F3: 184.8ms, SE = 2.3ms). While this negative lag was statistically significant ($t(515) = -3.00$, $p = 0.003$, 95% CI [-18.6, -3.9]), the effect size was negligible (Cohen's $d = -0.132$).

This modest overall effect, however, masks significant condition-specific differences that align with visual impressions. The timing relationship was strongly modulated by syllable position. Coda positions, where visual differences were most salient, showed that F3 peaked before articulation, with F3 for ɪ.coda peaking 29.8ms before articulation (SE = 6.1ms) and for æ.coda peaking 12.5ms before articulation (SE = 6.7ms). Conversely, onset positions displayed reversed patterns, with F3 for ɪ.onset peaking 3.9ms after articulation (SE = 8.8ms) and for æ.onset peaking 4.6ms after articulation (SE = 8.9ms).

The 33.7ms difference between ɪ.coda (-29.8ms) and æ.onset (+4.6ms) represents the most extreme temporal dissociation observed. The visual prominence of the coda effect, particularly for ɪ.coda with its early F3 peak around 150ms contrasting with articulatory peaks near 200ms, reflects this condition-specific pattern rather than the averaged overall effect. The differing timing relationships between F3 and articulatory peaks across coda and onset positions suggest different articulatory-acoustic coupling strategies. This pattern for F3 contrasts sharply with the consistent positive lags shown by F1 and F2, suggesting F3′s unique role as an anticipatory acoustic cue of lateral articulation.

3.3. Articulatory-acoustic relationships

3.3.1. Tongue lateralization (ΔHeight) and F1 analysis

Table 2 presents the results of the linear mixed-effects model examining the relationship between normalized F1 frequency and ΔHeight (tongue lateralization). The model revealed significant effects indicating vowel-specific patterns in the relationship between tongue lateralization and F1 frequency.

A significant main effect of ΔHeight was observed ($\beta = -0.064$, SE = 0.016, $p < 0.001$), showing an overall negative relationship between lateralization and F1 frequency across both vowels. The main effect is conditioned by the significant ΔHeight × vowel interaction ($\beta = 0.090$, SE = 0.023, $p < 0.001$), which reveals that the relationship between lateralization and F1 differs substantially between /æ/ and /ɪ/.

Table 2
Results of the linear mixed-effects (LME) model on the relationship between normalized F1 frequency and ΔHeight (tongue lateralization), with fixed effects of vowel (/ɪ/ and /æ/), and syllable position (onset and coda), and random effects for speaker, item, and repetition of the same token. The results in the model summary are for treatment contrasts. /æ/ is the referent vowel, and coda /l/ is the referent syllable position. Significant effects are shown in boldface.

Main Effects and Interactions	β	S.E.	t value	Pr (>|t|)
Intercept	0.273	0.132	2.068	0.076
ΔHeight (tongue lateralization)	-0.064	0.014	-4.632	**<0.001**
vowel	-0.688	0.085	-8.070	**<0.001**
syllable position	-0.047	0.077	-0.611	0.551
ΔHeight*vowel	0.090	0.020	4.511	**<0.001**
ΔHeight*syllable position	-0.008	0.022	-0.341	0.733
vowel*syllable position	0.064	0.126	0.506	0.628
ΔHeight*vowel*syllable position	0.042	0.033	1.300	0.194

(Bonferroni-adjusted significance level = 0.017)

For /æ/ (the reference level in the model), the relationship between ΔHeight and F1 follows the main effect slope of -0.064, indicating that increased lateralization (positive ΔHeight values, where the dominant tongue side is positioned lower relative to the midline) corresponds to decreased F1 frequency. For /ɪ/, the relationship is markedly different, with an effective slope of +0.026 (calculated as -0.064 + 0.090). This positive relationship indicates that increased lateralization corresponds to increased F1 frequency for /ɪ/, directly opposing the pattern observed for /æ/.

The vowel main effect was highly significant ($\beta = -0.688$, SE = 0.055, $p < 0.001$), confirming that F1 frequency is substantially lower for /ɪ/ compared to /æ/, consistent with the expected acoustic differences between these vowels. Both the main effect of syllable position ($\beta = -0.047$, SE = 0.078, $p = 0.551$) and its interaction with ΔHeight (($\beta = -0.008$, SE = 0.022, $p = 0.733$) were non-significant, providing no evidence that syllable position modulates the effect of lateralization on F1.

Fig. 10 provides clear visual confirmation of these statistical results. The scatter plot shows F1 frequency plotted against ΔHeight for both vowels in onset and coda positions. Consistent with the model results, the data points for both vowels cluster predominantly around ΔHeight = 0, reflecting minimal lateralization in most productions. The fitted regression lines in Fig. 10 clearly demonstrate the vowel-specific patterns identified in the statistical analysis. The red line representing /æ/ shows a negative slope, visualizing the inverse relationship between lateralization and F1 (slope = -0.064). The blue line for /ɪ/ displays a positive slope, confirming the opposite relationship for this vowel (slope = +0.026). The regression lines maintain consistent slopes across both onset and coda positions (as evidenced by the non-significant ΔHeight × syllable position interaction), confirming that the vowel-specific lateralization effects operate similarly regardless of syllable position.

3.3.2. Tongue lateralization (ΔHeight) and F2 analysis

Table 3 presents the results of the linear mixed-effects model examining the relationship between normalized F2 frequency and ΔHeight (tongue lateralization). The model revealed a complex pattern of significant effects and interactions that demonstrate both vowel-specific and position-specific patterns in the relationship between tongue lateralization and F2 frequency.

A significant main effect of ΔHeight was observed ($\beta = 0.040$, SE = 0.016, $p < 0.05$), indicating an overall positive relationship between lateralization and F2 frequency in the reference condition (/æ/ in coda position). However, this main effect should be interpreted in light of multiple significant interactions that reveal substantial context-dependent variation in the lateralization-F2 relationship.

The vowel main effect was not significant after Bonferroni correction ($\beta = 0.188$, SE = 0.063, $p = 0.023 > 0.017$). In contrast, the syllable position main effect was significant ($\beta = 0.233$, SE = 0.065, $p < 0.01$), suggesting that F2 frequency differs between onset and coda contexts regardless of vowel identity.

The ΔHeight × vowel interaction was significant ($\beta = 0.070$, SE = 0.023, $p < 0.01$), indicating that the relationship between lateralization and F2 differs between /æ/ and /ɪ/. Additionally, the ΔHeight × syllable interaction was significant ($\beta = -0.066$, SE = 0.025, $p < 0.05$), revealing that syllable position modulates the lateralization-F2 relationship.The three-way ΔHeight × vowel × syllable interaction was a key finding ($\beta = -0.130$, SE = 0.038, $p < 0.001$), showing that the combined effects of vowel context and syllable position create distinct patterns that cannot be predicted from their individual effects alone.

The significant interactions reveal four distinct patterns corresponding to each vowel-position combination. For /æ/ in coda /l/ position (the reference condition), the relationship between ΔHeight and F2 follows the main effect slope of +0.040, indicating that increased lateralization corresponds to increased F2 frequency.

For /ɪ/ in coda position, the relationship is enhanced with an effective slope of +0.110 (calculated as 0.040 + 0.070), showing a stronger positive relationship between lateralization and F2 frequency

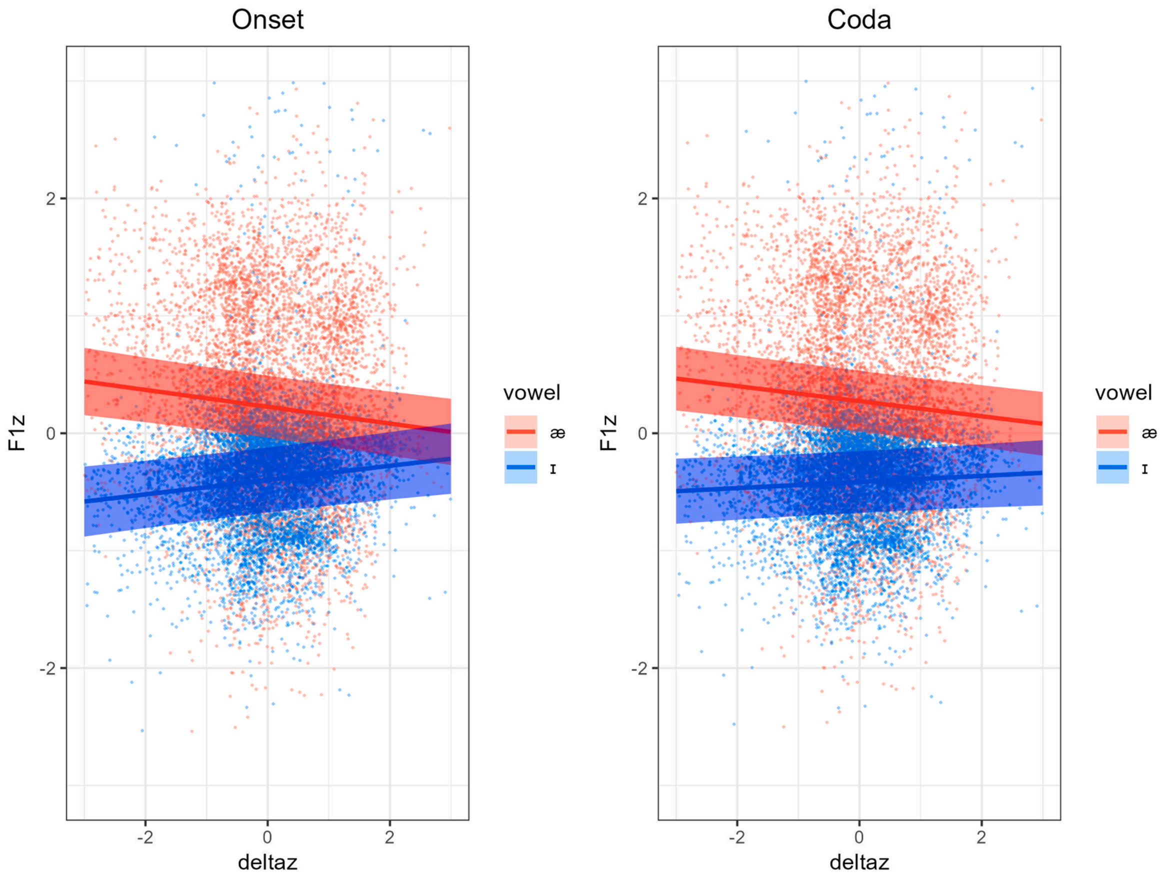

Fig. 10. (Color online) The relation of F1 frequency and ΔHeight (tongue lateralization). Data points included here are those that were equal to or less than three standard deviations from model predictions. The x-axis shows ΔHeight; the y-axis shows F1 frequency. Vowel context is separated by color and syllable position is separated by plots (left panel: onset position; right panel: coda position).

Table 3
Results of the linear mixed-effects (LME) model on the relationship between normalized F2 frequency and ΔHeight. The fixed effects were vowel (/ɪ/ and /æ/), and syllable position (onset and coda). Random effects were included for speaker, item and repetition of the same token. Significant effects are shown in boldface.

Main Effects and Interactions	β	S.E.	t value	Pr (>|t|)
Intercept	-0.292	0.132	-2.215	0.067
ΔHeight	0.040	0.016	2.467	**0.013**
vowel	0.188	0.063	2.947	0.023
syllable position	0.233	0.065	3.609	**0.004**
ΔHeight*vowel	0.070	0.023	3.038	**0.002**
ΔHeight*syllable position	-0.066	0.025	-2.593	**0.010**
vowel*syllable position	0.102	0.097	1.053	0.324
ΔHeight*vowel*syllable position	-0.130	0.038	-3.483	**<0.001**

(Bonferroni-adjusted significance level = 0.017)

compared to /æ/.

The pattern reverses in onset position. For /æ/ in onset position, the effective slope becomes -0.026 (calculated as 0.040 + (-0.066)), indicating a slight negative relationship between lateralization and F2. For /ɪ/ in onset position, this negative relationship is amplified with an effective slope of -0.086 (calculated as 0.040 + 0.070 + (-0.066) + (-0.130)), showing the strongest negative relationship among all conditions.

Fig. 11 provides clear visual confirmation of these statistical results. The scatter plots show F2 frequency plotted against ΔHeight for both vowels, separated by syllable position (onset and coda). Consistent with the model results, the data points cluster predominantly around ΔHeight = 0, reflecting minimal lateralization in most productions.

The fitted regression lines in Fig. 11 clearly demonstrate the context-specific patterns identified in the statistical analysis. In the coda panel, both the red line representing /æ/ and the blue line for /ɪ/ show positive slopes, with /ɪ/ displaying a steeper upward trajectory (slope = +0.110) compared to /æ/ (slope = +0.040). In the onset panel, both lines show negative slopes, with /ɪ/ again showing a steeper downward trajectory (slope = -0.086) compared to /æ/ (slope = -0.026).

3.3.3. *Tongue lateralization (ΔHeight) and F3 analysis*

Table 4 presents the results of the linear mixed-effects model examining the relationship between normalized F3 frequency and ΔHeight (tongue lateralization). The model revealed a distinct pattern from the F2 analysis, with fewer significant interactions but clear main effects that demonstrate both vowel-specific and context-dependent patterns in the relationship between tongue lateralization and F3 frequency.

A highly significant main effect of ΔHeight was observed (β = 0.081, SE = 0.016, $p < 0.001$), indicating a strong positive relationship between lateralization and F3 frequency across all conditions. This represents the most robust lateralization effect observed across all formant analyses.

The vowel main effect was significant (β = 0.263, SE = 0.062, $p < 0.01$), confirming that F3 frequency is significantly higher for /ɪ/ compared to /æ/, consistent with the expected acoustic differences between these vowels (Hillenbrand et al., 1995). Syllable position showed no significant main effect (β = -0.057, SE = 0.066, p = 0.414), indicating that overall F3 frequency does not differ systematically between onset and coda positions.

Neither the ΔHeight × vowel interaction (β = 0.005, SE = 0.023, p = 0.819) nor the ΔHeight × syllable interaction (β = 0.028, SE = 0.025, p

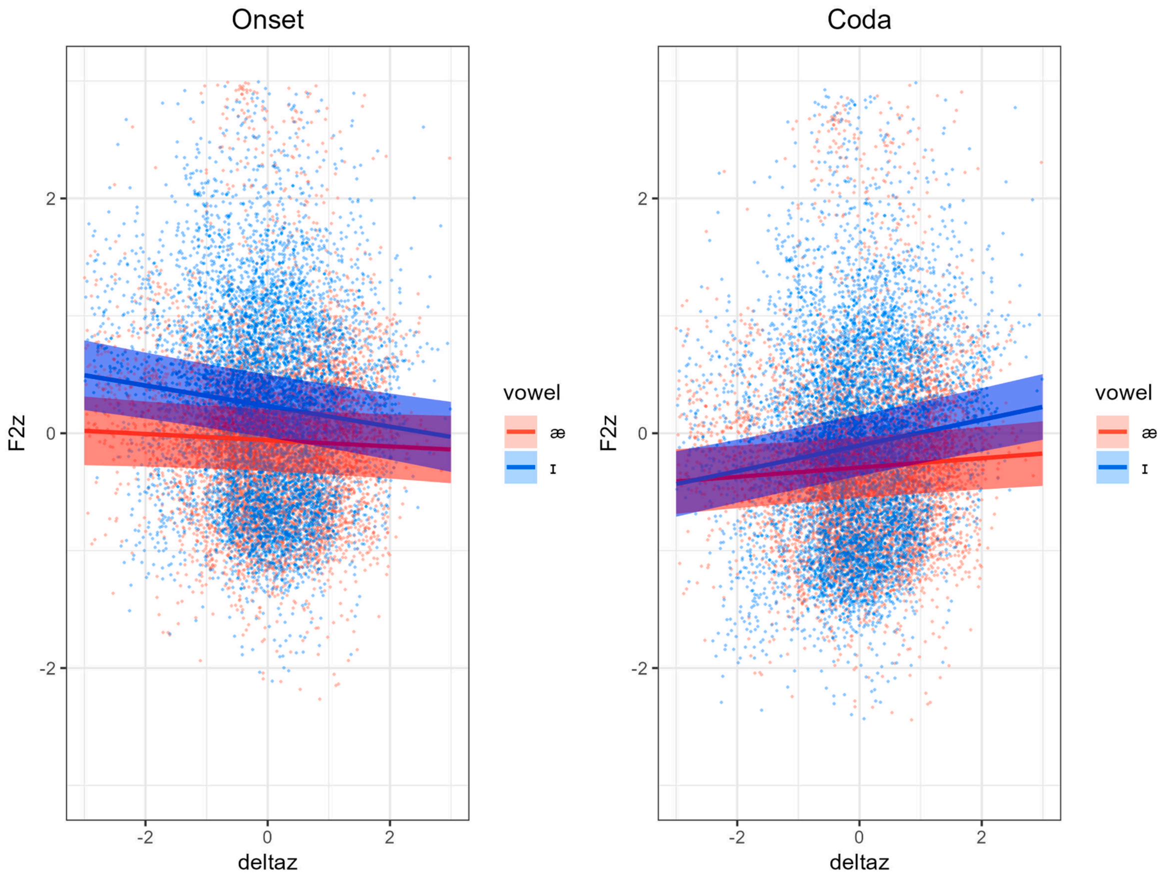

Fig. 11. (Color online) The relation of normalized F2 frequency and ΔHeight. Data points included here are those that were equal to or less than three standard deviations from model predictions. The x-axis shows ΔHeight; the y-axis shows F2 frequency. Vowel context is separated by color and syllable position is separated by plots (left panel: onset position; right panel: coda position).

Table 4
Results of the linear mixed-effects (LME) model on the relationship between normalized F3 frequency and ΔHeight. The fixed effects were vowel (/ɪ/ and /æ/), and syllable position (onset and coda). Random effects were included for speaker, item and repetition of the same token. Significant effects are shown in boldface.

Main Effects and Interactions	β	S.E.	t value	Pr (>|t|)
Intercept	-0.133	0.303	-0.440	0.678
ΔHeight	0.081	0.016	5.115	**<0.001**
vowel	0.263	0.062	4.217	**0.004**
syllable position	-0.057	0.066	-0.857	0.414
ΔHeight*vowel	0.005	0.023	0.229	0.819
ΔHeight*syllable position	0.028	0.025	1.104	0.270
vowel*syllable position	0.048	0.097	0.496	0.634
ΔHeight*vowel*syllable position	0.103	0.037	2.754	**0.006**

(Bonferroni-adjusted significance level = 0.017)

= 0.270) reached significance, suggesting that the basic lateralization-F3 relationship is relatively consistent across vowel types and syllable positions. However, the three-way ΔHeight × vowel × syllable interaction was significant ($\beta = 0.103$, $SE = 0.037$, $p < 0.01$), indicating that a specific combination of vowel identity and syllable position does modulate the lateralization effect.

For most conditions (/æ/ in both positions and /ɪ/ in coda position), the relationship between ΔHeight and F3 follows the main effect slope of approximately +0.081, indicating that increased lateralization corresponds to increased F3 frequency.

However, for /ɪ/ in onset position, the relationship is enhanced with an effective slope of +0.217 (calculated as 0.081 + 0.005 + 0.028 + 0.103), showing a substantially stronger positive relationship between lateralization and F3 frequency compared to all other conditions.

Fig. 12 provides clear visual confirmation of these statistical results. The scatter plots show F3 frequency plotted against ΔHeight for both vowels, separated by syllable position (onset and coda). Consistent with previous analyses, the data points cluster predominantly around ΔHeight = 0, reflecting minimal lateralization in most productions.

The fitted regression lines in Fig. 12 clearly demonstrate the context-specific patterns identified in the statistical analysis. In the coda panel, both the red line representing /æ/ and the blue line for /ɪ/ show similar positive slopes, consistent with the non-significant two-way interactions. In the onset panel, the /æ/ line maintains a similar positive slope to the coda conditions, while the /ɪ/ line shows a steeper positive slope, visually confirming the enhanced lateralization effect for /ɪ/ in onset position identified in the three-way interaction.

3.4. Relative importance analysis

Given that ΔHeight (tongue lateralization) can be used as a predictor of F1, F2 and F3, a relative importance analysis was performed. In this case, ΔHeight was modelled using the formants (F1, F2, and F3 frequencies). The following model was tested: lm (ΔHeight ~ F1 + F2 + F3). The proportion of variance explained by the model is about 6.32%. The metrics are normalized to sum to 100%. The relative importance metrics are: 27% for F1 frequency, 21% for F2 frequency, and 52% for F3 frequency. As mentioned earlier, a predictor with larger percentage means that it is more important in predicting the values of the relevant factor. Therefore, the relative importance analysis indicates that ΔHeight predicts F3 to a greater degree than it predicts F1, and predicts F1 to a somewhat greater degree than F2.

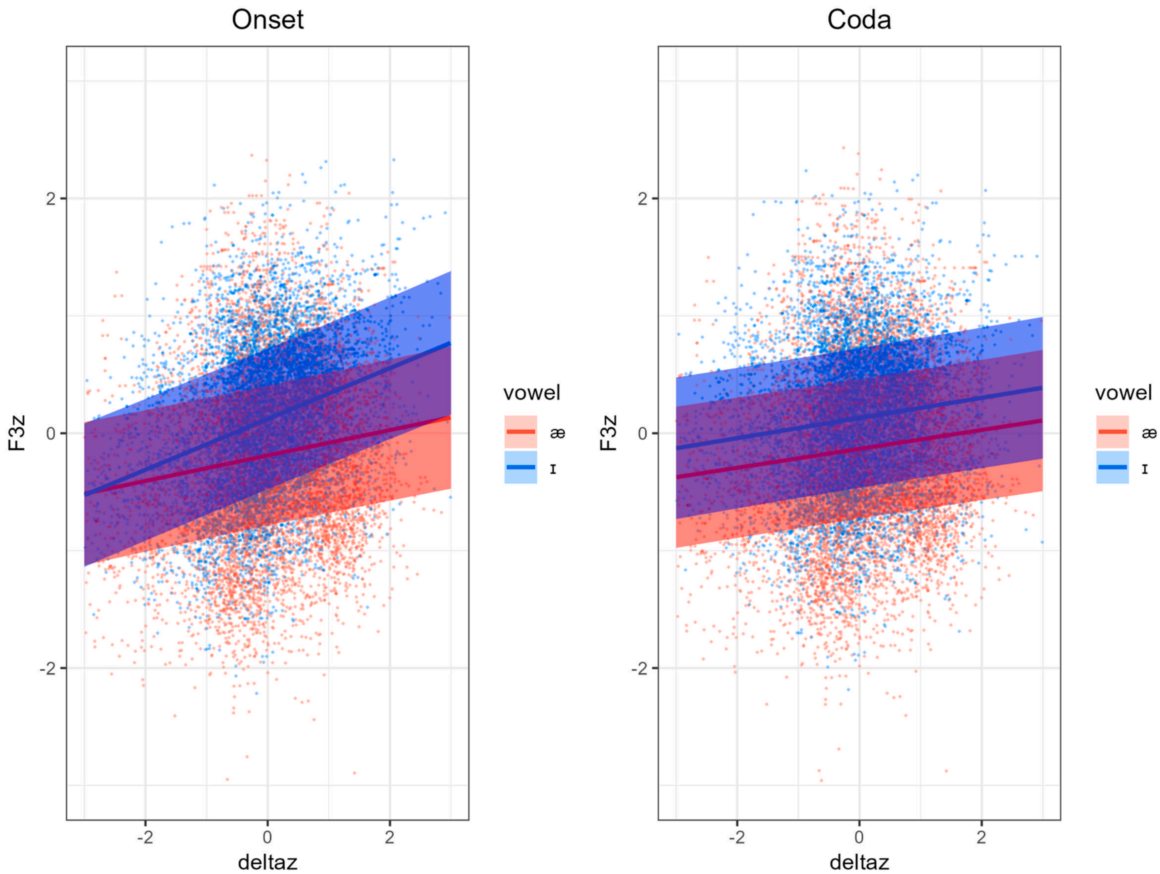

Fig. 12. (Color online) The relation of normalized F3 frequency and ΔHeight. Data points included are those that were equal to or less than three standard deviations from model predictions. The x-axis shows ΔHeight; the y-axis shows F3 frequency. Vowel context is separated by color and syllable position is separated by plots (left panel: onset position; right panel: coda position).

4. Discussion

The present study employed two analytical approaches that yielded contrasting patterns. The peak timing analysis examined temporal coordination between articulatory and acoustic events, revealing significant syllable position effects. The trajectory analysis (LME models) examined how formant frequencies change with lateralization degree, revealing primarily vowel effects. These analyses capture fundamentally different aspects of lateral production: timing analysis reflects gestural coordination, while trajectory analysis reflects articulatory-acoustic coupling. Given that the hypotheses specifically address the relationship between lateralization degree and formant frequencies, this analysis focuses on the trajectory analysis results for hypothesis evaluation.

Hypothesis 1 was fully supported. As predicted, /ɪl/ showed significant F1 increases with greater lateralization ($p < .05$), consistent with blade lowering during lateralization. Crucially, /æl/ showed significant F1 decreases with lateralization ($p < .05$), supporting the proposal that speakers employ a compensatory raising strategy when transitioning from low /æ/ to high /l/. These opposing patterns demonstrate that lateralization is not a uniform articulatory gesture but rather involves context-dependent strategies shaped by the articulatory demands of the vowel-to-/l/ transition. The results challenge traditional assumptions about lateralization and highlight the complex interaction between segmental targets and articulatory implementation (Sproat and Fujimra, 1993).

Hypothesis 2 was not supported. Contrary to predictions, F2 showed significant correlations with lateralization. Moreover, F1 and F2 demonstrated opposing patterns. F1 was sensitive to vowel context but not syllable position, whereas F2 was sensitive to syllable position ($p < 0.01$) but not to vowel context. This position-dependent pattern, along with the complex interactions, suggests that F2 reflects different articulatory dimensions than F1, possibly related to tongue body positioning that varies with syllable position rather than vowel identity.

Hypothesis 3 (F3 primacy) was strongly supported. F3 showed the strongest and most consistent positive correlation with lateralization across all conditions (main effect: $\beta = 0.081$, $p < 0.001$). Among the three formants, F3 accounted for 52% of the predictive variance (compared to F1: 27% and F2: 21%), establishing it as the dominant acoustic correlate. The significant vowel main effect ($\beta = 0.263$, $p < 0.01$) indicates that /ɪ/ has inherently higher F3 than /æ/, consistent with known vowel acoustics. Most notably, the three-way interaction ($\beta = 0.103$, $p < 0.01$) revealed that /ɪ/ in onset position shows an enhanced lateralization effect (slope $= +0.217$) compared to all other conditions (slope $\approx +0.081$). The positive slope in /ɪ/-onset position suggests that baseline vocal tract configuration modulates how strongly channel asymmetry affects F3. In high front vowels like /ɪ/, F3 is already elevated, potentially placing it closer to the frequency region where lateral channel effects are strongest.

Hypothesis 4 (position stability) was fully supported. As predicted, F1 and F3 showed no significant syllable position main effects on the lateralization-formant relationship, confirming that these lateralization-related formants remain stable across positions. In contrast, F2 showed significant position effects ($\beta = 0.233$, $p < 0.01$) and position-dependent interactions, exactly as predicted for a formant that reflects tongue body retraction and timing rather than lateralization. This pattern confirms that while the degree of lateralization remains stable across syllable positions (reflected in consistent F1 and F3 patterns), other aspects of /l/ articulation such as tongue body positioning vary systematically between onset and coda positions (reflected in F2). The results provide strong support for distinguishing lateralization-related acoustic effects from positional variations in tongue body dynamics.

Vowel Context Dominance: A systematic pattern emerging across the analyses reveals that vowel effects exceed lateralization effects for all formants (F1: βvowel = -0.688 vs. βΔHeight = -0.064; F2: βvowel = 0.188 vs. βΔHeight = 0.040; F3: βvowel = 0.263 vs. βΔHeight = 0.081). This dominance of vowel context indicates that /l/ acoustics are not simply the product of an invariant lateral gesture but emerge from the dynamic interaction between vowel-specific articulatory configurations and lateral channel formation. The significant vowel-lateralization interactions for F1 (β = 0.090) and F2 (β = 0.070) provide further evidence that lateralization has context-dependent rather than uniform acoustic consequences. From a theoretical perspective, these findings align with models emphasizing overlapping articulatory gestures (Fowler and Saltzman, 1993; Goldstein and Fowler, 2003), where lateral channel formation operates as a semi-independent articulatory dimension that modulates but does not override vowel-specific configurations.

F3 as the Primary Acoustic Correlate: F3 has been established as the dominant acoustic correlate of lateralization (Espy-Wilson, 1992; Stevens, 1998; Charles and Lulich, 2019). This finding requires an investigation into the underlying mechanisms that drive this effect. The strong positive correlation between lateralization and F3 frequency provides empirical evidence for theoretical models predicting that lateral channel formation introduces complex cavity interactions (Narayanan et al., 1997; Zhang and Espy-Wilson, 2004). During /l/ production, asymmetric tongue blade positioning creates lateral channels with different cross-sectional areas—the elevated side forms a more constricted channel while the lowered side creates a less constricted pathway (Narayanan et al., 1997; Charles and Lulich, 2019). This area-based asymmetry generates different acoustic impedances and resonance characteristics between channels.

The acoustic effects may arise through multiple mechanisms: direct spectral energy contribution from side branch resonances, pole-zero interactions where lateral cavities introduce anti-formants that suppress the original F3 (Bangayan et al., 1996; Narayanan and Alwan, 1996; Zhang and Espy-Wilson, 2004; Charles and Lulich, 2019), or complex interactions between the main cavity, supralingual cavity, and sublingual spaces (Narayanan et al., 1997). Charles and Lulich (2019) found that anti-resonances from the supralingual cavity are more prominent than those from interdental channels, though the latter can become important in asymmetrical articulations.

Temporal Dynamics of Articulatory-Acoustic Coupling: The temporal relationship between articulatory lateralization peaks and acoustic formant peaks reveals formant-specific coupling mechanisms that illuminate the dynamic nature of /l/ production. F3 demonstrated a qualitatively different pattern from F1 and F2, with acoustic peaks preceding articulatory peaks by an average of 11.2 ms, most pronounced in the /ɪ/ + coda condition (-29.8 ms). This anticipatory acoustic behavior suggests that F3 captures early acoustic consequences of the developing lateral configuration as the articulators approach their target positions.

In contrast, F1 acoustic peaks lagged behind articulatory peaks (13.8 ms), indicating they reflect the acoustic consequences of achieved configurations rather than anticipatory adjustments. F2 showed limited analyzable peaks, with only the æ.coda condition displaying a measurable peak that lagged 10.1 ms behind articulation. The onset /l/s showed substantially larger lags compared to coda /l/s for F1 (onset: ~25 ms, coda: ~6 ms), suggesting that anticipatory coarticulation in onset position creates greater temporal dissociation between articulatory gesture and acoustic realization. The temporal precedence of F3 changes, combined with its strong magnitude correlation, establishes F3 as an early acoustic cue of lateral articulation that encodes anticipatory articulatory processes intrinsic to lateral production.

Implications for Understanding Lateral Articulation: These findings reveal that /l/ production involves temporally distributed articulatory adjustments with formant-specific acoustic consequences, challenging models that assume uniform articulatory-acoustic mapping (Sproat and Fujimura, 1993 . The relationship between F3 and lateralization is dynamic, defined by its anticipatory timing, consistent magnitude correlations, and context-dependent modulation. Together, these features demonstrate that lateral articulation operates through controlled, asymmetric channel formation with systematic acoustic encoding. Rather than passive consequences of tongue body movement, lateral channels function as active articulatory targets with dedicated acoustic correlates, supporting Sproat and Fujimura's (1993) proposal of active tongue tip and blade control during /l/ production. The results establish F3 as the primary acoustic dimension for encoding lateral gesture dynamics, providing a foundation for understanding how speakers control and coordinate the complex articulatory-acoustic relationships that define lateral consonant production across varying different contexts. While individual anatomical differences likely contribute additional variability (Lammert et al., 2013; Brunner et al., 2009; Fuchs et al., 2006), fundamental F3-lateralization coupling patterns nonetheless emerge consistently across speakers and contexts.

Limitations: The findings of this study should be considered in light of its methodological and analytical limitations. First, the virtual mid-sagittal tongue blade point (vTB) used to quantify lateralization is derived from three mid-sagittal EMA sensors without external imaging validation for the participants. The analysis relied on quadratic interpolation to estimate tongue curvature from three sensor points. This approach is standard in the field and represents the most complex model mathematically derivable from three points. However, a key limitation is that this method cannot capture more complex tongue shapes. Future work should incorporate synchronized ultrasound to acquire full tongue surface contours. This would enable the quantification of the interpolation error and an assessment of its potential influence on the timing relationships reported in this study. Real-time MRI, while informative for volumetric aspects, is presently impractical due to access and cost; ultrasound provides a feasible and information-rich next step.

Second, the linear mixed-effects models explained a relatively small proportion of variance (adjusted $R^2 = 1.98\%$), indicating that phonological context accounts for only a modest portion of total variability in lateralization-formant relationships. This finding is common in speech production research, as articulatory data inherently exhibits variability from multiple simultaneous factors: individual anatomical differences, variations in speaking rate and prosodic emphasis, measurement noise, and the complex multi-dimensional nature of speech motor control. Additionally, tongue movements are influenced by numerous unmeasured factors including coarticulatory effects from adjacent segments, and lexical frequency. While the models capture statistically reliable and linguistically meaningful patterns, the modest variance explained suggests the need for future research to incorporate additional predictors and potentially employ more sophisticated modeling approaches to better account for the multifactorial nature of speech production variability.

Future Directions: The findings raise several critical questions for future research. First, it remains crucial to identify and model the specific oral cavity modifications that drive anticipatory F3 changes during the approach to lateral constriction. High-resolution imaging (MRI/ultrasound) synchronized with acoustics could map the progression of cavity formation to F3 modulation. Second, the underlying acoustic mechanism requires verification. Future studies should determine whether the positive F3 shift reflects side branch resonance contributions, F3 suppression with F4 emergence, or both. Detailed spectral analysis examining F4 and F5 behavior during lateralization might distinguish these mechanisms. Third, context-specific pole-zero mapping could explain why /ɪ/ + onset shows enhanced F3 modulation; acoustic modeling with realistic vocal tract geometries could identify the cavity configurations that optimize pole-zero interactions. Finally, cross-linguistic generalization remains untested: does positive F3 modulation with lateralization occur universally, or is it specific to English /l/? Comparative studies with languages featuring different lateral types could test the generality of our findings. These investigations would clarify whether the positive lateralization-F3 relationship represents a universal acoustic consequence of lateral channel asymmetry or a language-specific pattern. This work would advance our understanding of how speakers achieve lateral acoustic goals through complex, context-dependent articulatory strategies.

5. Conclusion

This study demonstrates a robust positive correlation between lateral tongue channel formation and third formant (F3) frequency during Australian English /l/ production. The results show that F3 increases systematically with the magnitude of lateralization across all speakers and contexts. Fine-grained temporal analysis of synchronized articulatory and acoustic data found that F3 modulation anticipates peak lateralization. The strength of this relationship varies systematically with phonological context. These findings advance our understanding of lateral articulation by demonstrating that asymmetric tongue configurations function as active articulatory targets with systematic acoustic correlates, rather than passive byproducts of tongue tip elevation, and dorsal retraction (Sproat and Fujimura, 1993; Browman and Goldstein, 1995). The consistent positive F3-lateralization relationship across speakers, despite individual anatomical differences, suggests a fundamental constraint on how lateral channels modulate vocal tract acoustics.

Author statement

Jia Ying is the sole author and conducted all aspects of this research. This work is based on a chapter from the author's thesis completed in 2020 at Western Sydney University. The original thesis supervisor approved publication of this work as a sole-authored manuscript. The author is currently affiliated with the Chinese University of Hong Kong, Shenzhen and prepared the final manuscript independently.

CRediT authorship contribution statement

Jia Ying: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The research was made possible through funding support from multiple institutions: a PhD Scholarship and Research Training Support (RTS) fund from the MARCS Institute at Western Sydney University; the Research Project for Research Initiation Funds for Newly Introduced Top-notch and Urgently-needed Talents of the Chinese University of Hong Kong, Shenzhen (grant numbers KQ002520 and 2024TC0111); and the HSS SRG1-应佳 (G10120230302) grant from the Chinese University of Hong Kong, Shenzhen. The author extends sincere gratitude to the PhD committee members, Christopher Carignan, Michael Proctor, Jason A. Shaw, Donald Derrick, and Catherine T. Best, for their expert guidance, insightful comments, and technical support throughout this project, which have significantly enhanced the quality of this work. Deep appreciation is also expressed to all participants who generously contributed their time and effort to this research, and to the anonymous reviewers for their invaluable constructive feedback. While every effort has been made to ensure the accuracy and completeness of this work, any remaining errors or omissions are the sole responsibility of the author.

Data availability

Data will be made available on request.

References

Adank, P., Smits, R., Van Hout, R., 2004. A comparison of vowel normalization procedures for language variation research. J. Acoust. Soc. Am. 116 (5), 3099–3107.
Alwan, A., Narayanan, S., Haker, K., 1997. Toward articulatory-acoustic models for liquid approximants based on MRI and EPG data. Part II. The rhotics. J. Acoust. Soc. Am. 101 (2), 1078–1089.
Blackwood Ximenes, A., Shaw, J.A., Carignan, C., 2017. A comparison of acoustic and articulatory methods for analyzing vowel differences across dialects: Data from American and Australian English. J. Acoust. Soc. Am. 142 (1), 363–377.
Bladon, R.A., Al-Bamerni, A., 1976. Coarticulation resistance of English /l/. J. Phon. 4, 135–150.
Brunner, J., Fuchs, S., Perrier, P., 2009. On the relationship between palate shape and articulatory behavior. J. Acoust. Soc. Am. 125 (6), 3936–3949.
Byrd, D., 2000. Articulatory vowel lengthening and coordination at phrasal junctures. Phonetica 57 (1), 3–16.
Charles, S., Lulich, S.M., 2018. Case study of Brazilian Portuguese laterals using a novel articulatory-acoustic methodology with 3D/4D ultrasound. Speech Commun. 103, 37–48.
Charles, S., Lulich, S.M., 2019. Articulatory-acoustic relations in the production of alveolar and palatal lateral sounds in Brazilian Portuguese. J. Acoust. Soc. Am. 145 (6), 3269–3288.
Cruttenden, A., 2008. Gimson's Pronunciation of English, Seventh Edit. Arnold, London.
Espy-Wilson, C.Y., 1992. Acoustic measures for linguistic features distinguishing the semivowels /w j r l/in American English. J. Acoust. Soc. Am. 92 (2), 736–757.
Fant, G., 1960. Acoustic Theory of Speech Production. Mouton, The Hague.
Fowler, C.A., Saltzman, E., 1993. Coordination and coarticulation in speech production. Lang. Speech 36 (2-3), 171–195.
Jonathan Harrington Fuchs, S., Perrier, P., Geng, C., Mooshammer, C., 2006. What role does the palate play in speech motor control? Insights from tongue kinematics for German alveolar obstruents. In: Marija Tabain (Ed.), Speech Production. Models, Phonetic Processes, and Techniques. Psychology Press, New York and Hove, pp. 149–164.
Gibert, G., Olsen, K.N., Leung, Y., Stevens, C.J., 2015. Transforming an embodied conversational agent into an efficient talking head: from keyframe-based animation to multimodal concatenation synthesis. Comput. Cogn. Sci. 1 (1), 1–12.
Gick, B., Allen, B., Roewer-Després, F., Stavness, I., 2017. Speaking tongues are actively braced. J. Speech Lang. Hear. Res. 60 (3), 494–506.
Goldstein, L., & Fowler, C. A. (2003). Articulatory phonology: a phonology for public language use. Phonetics and phonology in language comprehension and production: Differences and similarities, 159, 207.
Grömping, U., 2006. Relative importance for linear regression in R: the package Relaimpo. J. Stat. Softw. 17 (1), 1–27.
Hillenbrand, J., Getty, L.A., Clark, M.J., Wheeler, K., 1995. Acoustic characteristics of American English vowels. J. Acoust. Soc. Am. 97 (5), 3099–3111.
Hwang, Y., Charles, S., Lulich, S.M., 2019. Articulatory characteristics and variation of Korean laterals. Phon. Speech Sci. 11 (1), 19–27.
Hwang, Y., Lulich, S.M., De Jong, K.J., 2019. Articulatory and acoustic characteristics of the Korean and English word-final laterals produced by Korean female learners of American English. J. Acoust. Soc. Am. 146 (5), EL444–EL450.
Kalashnikova, M., Carignan, C., Burnham, D., 2017. The origins of babytalk: smiling, teaching or social convergence? R. Soc. Open Sci. 4 (8), 170306.
Katz, W.F., Bharadwaj, S.V., Stettler, M.P., 2006. Influences of electromagnetic articulography sensors on speech produced by healthy adults and individuals with aphasia and apraxia. JSLHR 49, 645–659.
Ladefoged, P., 2001. Vowels and consonants. Phonetica 58 (3), 211–212.
Ladefoged, P., Johnson, K., 2014. A Course in Phonetics, 7th ed. Cengage Learning. Inc., Boston.
Ladefoged, P., Maddieson, I., 1996. The Sounds of the World's Languages. Blackwell Publishing, Malden, MA (USA).
Lammert, A., Proctor, M., Narayanan, S., 2013. Interspeaker variability in hard palate morphology and vowel production. JSLHR 56 (6), 1924–1933.
Lehiste, I. (1964). "Acoustical characteristics of selected English consonants". Indiana research center in Anthropology, Folklore and Linguistics, 34, 10–50.
Lin, S.S., 2011. Production and Perception of Prosodically Varying Inter-Gestural Timing in American English Laterals. University of Michigan.
Lindeman, R.H., Merenda, P.F., Gold, R.Z., 1980. Introduction to Bivariate and Multivariate Analysis, 4. Scott, Foresman, Glenview, IL.
Narayanan, S., Alwan, A., 1996. Parametric hybrid source models for voiced and voiceless fricative consonants. In: Proceedings of the International Conference on Acoustics, Speech, and Signal Processing. Atlanta, Georgia, 1, pp. 377–380. May 7-10.
Narayanan, S.S., Alwan, A.A., Haker, K., 1997. Toward articulatory-acoustic models for liquid approximants based on MRI and EPG data. Part I. The laterals. J. Acoust. Soc. Am. 101 (2), 1064–1077.
Narayanan, S., Byrd, D., Kaun, A., 1999. Geometry, kinematics, and acoustics of Tamil liquid consonants. J. Acoust. Soc. Am. 106 (4), 1993–2007.

Proctor, M., 2011. Towards a gestural characterization of liquids: evidence from Spanish and Russian. LabPhon 2 (2), 451–485.
Sproat, R., Fujimura, O., 1993. Allophonic variation in English /l/and its implications for phonetic implementation. J. Phon. 21 (3), 291–311.
Stevens, K., 1998. Acoustic Phonetics. The MIT Press, Cambridge MA.
Stone, M., Faber, A., Cordaro, M., 1991. Cross-sectional tongue movement and tongue-palate movement in [s] and [ʃ] syllables. In: Proceedings of the 13th International Congress of Phonetic Sciences. Stockholm, Sweden, 2, pp. 354–357. August 19-24.
Stone, M., Faber, A., Raphael, L., Shawker, T., 1992. "Cross-sectional tongue shapes and linguopalatal contact patterns in [s], [ʃ] and [l] syllables. J. Phon. 20, 253–270.
Stone, M., Lundberg, A., 1996. Three-dimensional tongue surface shapes of English consonants and vowels. J. Acoust. Soc. Am. 99 (6), 3728–3737.
Strycharczuk, P., Derrick, D., & Shaw, J. (2020). Locating de-lateralization in the pathway of sound changes affecting coda/l.
Szalay, T., Proctor, M., Gully, A., Piyadasa, T., Jin, C., Waddington, D., Ballard, K., 2024. August). Lateral articulation across vowel contexts: insights from Magnetic Resonance Imaging. In: Proceedings of the 19th Australasian International Conference on Speech Science and Technology (SST2024). York.
Turton, D., 2017. Categorical or gradient? An ultrasound investigation of /l/-darkening and vocalization in varieties of English. Lab. Phonol. 8 (1).
Wieling, M., Tomaschek, F., Arnold, D., Tiede, M., Bröker, F., Thiele, S., Wood, S.N., Baayen, R.H., 2016. Investigating dialectal differences using articulography. J. Phon. 59, 122–143.
Wrench, A.A., Scobbie, J.M., 2003. Categorising vocalisation of English/l/using EPG, EMA and ultrasound. In: Proceedings of the 6th International Seminar on Speech Production.
Ying, J., Shaw, J.A., Carignan, C., Proctor, M., Derrick, D., Best, C.T., 2021. Evidence for active control of tongue lateralization in Australian English /l/. J. Phon. 86, 101039.
Zhang, Z., Espy-Wilson, C.Y., 2004. A vocal-tract model of American English /l/. J. Acoust. Soc. Am. 115 (3), 1274–1280.